Layers

- # PLMN Infographic
- Frame 1 - What is a PLMN
  - Header
  - Definition card
  - PLMN ID blocks
- Frame 2 - Architecture
  - RAN / Core diagram
  - Node table
- Frame 3 - Design Prototype
  - Workflow cards
  - Roaming diagram

Frame 1 - What is a PLMN



1654 x 2339

Frame 2 - Architecture



1654 x 2339

Frame 3 - Design Prototype



1654 x 2339

1 | DESIGN
Layout, colour styles, type and components

Design

Color styles

Navy, Blue, Cyan, Teal
Orange, Purple, Pink, Green

Text styles

Title / Bold 76

Heading / Bold 40

Body / Regular 30

Caption / 26

Layout grid

Frame: 1654 x 2339 (A4)
Margin: 80 px
Corner radius: 24-36 px
Shadow: Y10, blur 14

Components

- Info card
- Step badge
- Header band
- Arrow connector

Flow starting point

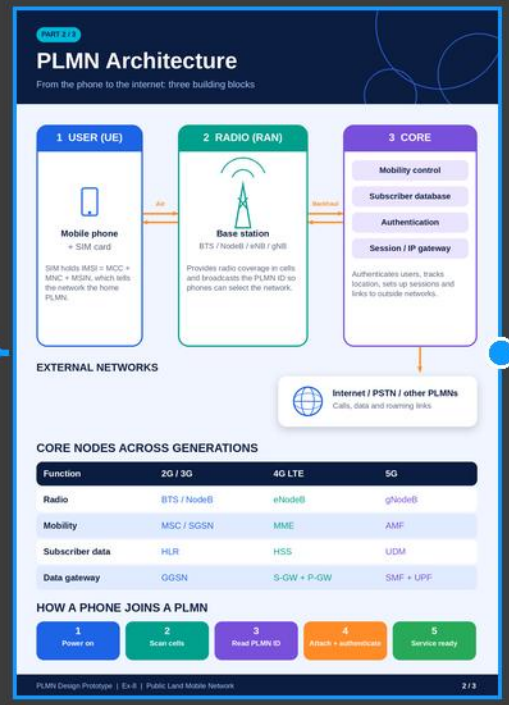
Frame 1 - What is a PLMN



Next >

On click

Frame 2 - Architecture



Next >

On click

Frame 3 - Design Prototype



Restart

2 | PROTOTYPE
Clickable flow: Frame 1 > Frame 2 > Frame 3

Prototype

Interaction

Trigger	On click
Action	Navigate to
Destination	Next frame
Animation	Smart animate
Easing	Ease out
Duration	300 ms
Device	Desktop / A4

Flow 1

1. What is a PLMN
2. Architecture
3. Design Prototype

Play >

FINAL OUTPUT PLMN infographic - three pages in order

1

PART 1 / 3

What is a PLMN?

Public Land Mobile Network - the operator network behind every mobile call

DEFINITION

A PLMN is a land-based wireless network operated by one mobile operator in one country, providing public mobile services (voice, SMS and data) to subscribers. Each PLMN is uniquely identified worldwide by a PLMN ID.

THE PLMN ID = MCC + MNC

MCC — Mobile Country Code - 3 digits

+

MNC — Mobile Network Code - 2 or 3 digits

Example (India): 404 - XX

India uses MCC 404 and 405. The MNC identifies the operator, so PLMN ID 404-XX means 'operator XX in India'. The same MCC+MNC prefixes the subscriber's IMSI.

KEY FACTS

1 **One operator** — Each PLMN belongs to a single operator in a single country.

2 **Two parts** — Radio access network (RAN) plus the core network.

3 **Many users** — Serves subscribers at home and visiting roamers.

SERVICES CARRIED BY A PLMN

Voice | SMS | Mobile data | Roaming

Remember: PLMN = who operates the network (MCC+MNC) and what it connects (RAN + core).

PLMN Design Prototype | Ex-8 | Public Land Mobile Network 1 / 3

What is a PLMN?

2

PART 2 / 3

PLMN Architecture

From the phone to the internet: three building blocks

1 USER (UE)

Mobile phone
+ SIM card

SIM holds IMSI = MCC + MNC + MSIN, which tells the network the home PLMN.

Air

2 RADIO (RAN)

Base station
BTS / NodeB / eNB / gNB

Provides radio coverage in cells and broadcasts the PLMN ID so phones can select the network.

Backhaul

3 CORE

- Mobility control
- Subscriber database
- Authentication
- Session / IP gateway

Authenticates users, tracks location, sets up sessions and links to outside networks.

EXTERNAL NETWORKS

Internet / PSTN / other PLMNs
Calls, data and roaming links

CORE NODES ACROSS GENERATIONS

Function	2G / 3G	4G LTE	5G
Radio	BTS / NodeB	eNodeB	gNodeB
Mobility	MSC / SGSN	MME	AMF
Subscriber data	HLR	HSS	UDM
Data gateway	GGSN	S-GW + P-GW	SMF + UPF

HOW A PHONE JOINS A PLMN

1 Power on → 2 Scan cells → 3 Read PLMN ID → 4 Attach + authenticate → 5 Service ready

PLMN Design Prototype | Ex-8 | Public Land Mobile Network 2 / 3

PLMN Architecture

3

PART 3 / 3

PLMN Design Prototype

Planning steps, PLMN types and roaming

DESIGN WORKFLOW

1 **Requirements** — Target area, subscribers, services and traffic forecast.

2 **PLMN ID & numbering** — Assign MCC+MNC, IMSI ranges, MSISDN plan.

3 **Spectrum & RF plan** — Choose bands, cell sites, coverage and capacity.

4 **Transport / backhaul** — Fibre or microwave links from sites to the core.

5 **Core design** — Mobility, subscriber data, gateways, security.

6 **Test & optimise** — Drive tests, KPIs, handover and roaming tests.

PLMN TYPES IN ROAMING

HPLMN — Home PLMN
Operator where the subscriber's SIM is registered.

VPLMN — Visited PLMN
Foreign or other network the user roams onto.

EPLMN — Equivalent PLMN
Networks treated like the home PLMN for selection.

Home PLMN (Core + subscriber data) ⇄ Visited PLMN (Local radio + core)
Authentication + billing
Roaming agreement

DESIGN CHECKLIST

- ✓ Unique PLMN ID (MCC+MNC)
- ✓ Coverage and capacity targets met
- ✓ Secure authentication (SIM / AKA)
- ✓ Roaming agreements in place

PLMN = Operator ID + Radio access + Core network + Roaming links

PLMN Design Prototype | Ex-8 | Public Land Mobile Network 3 / 3

PLMN Design Prototype